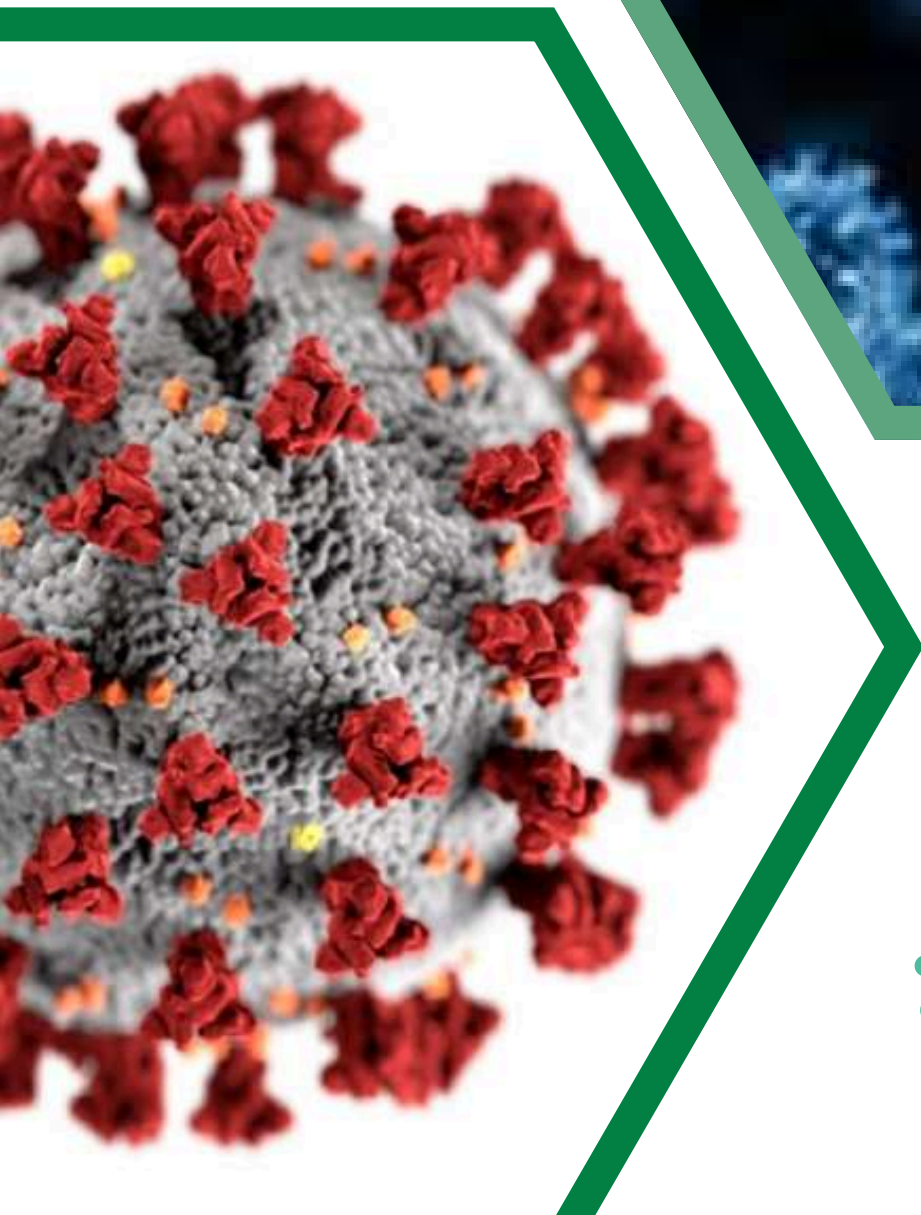
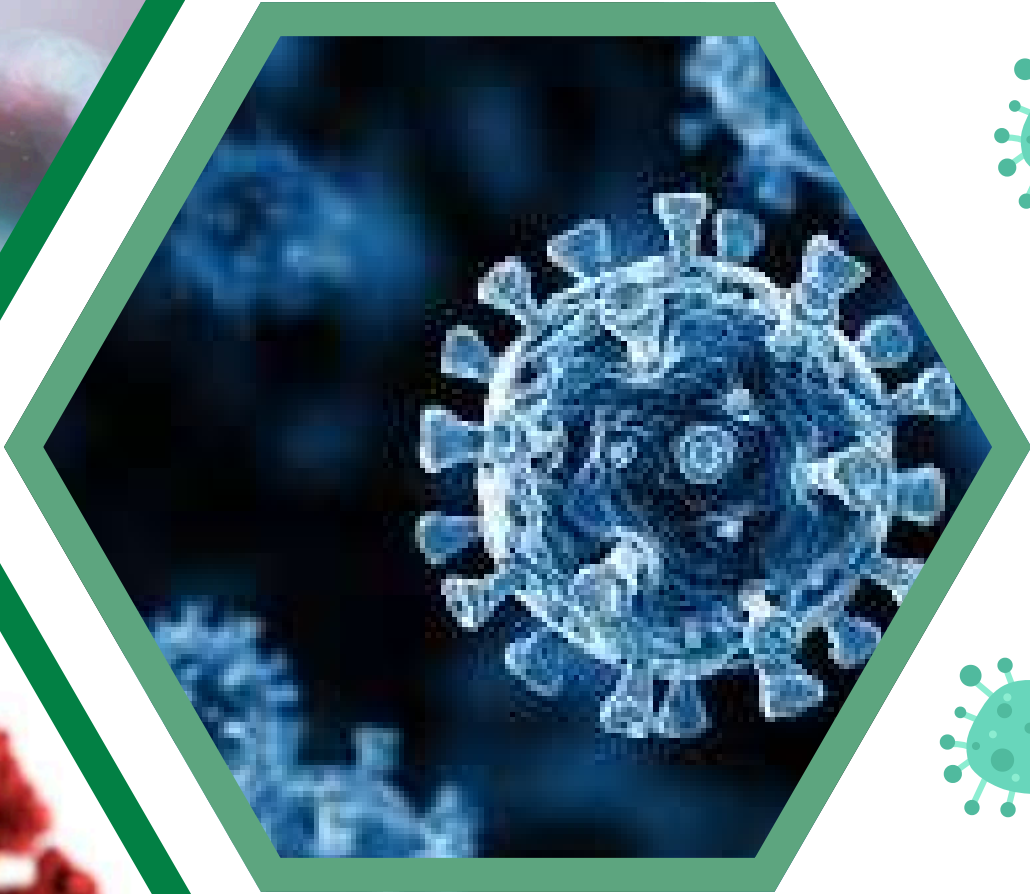
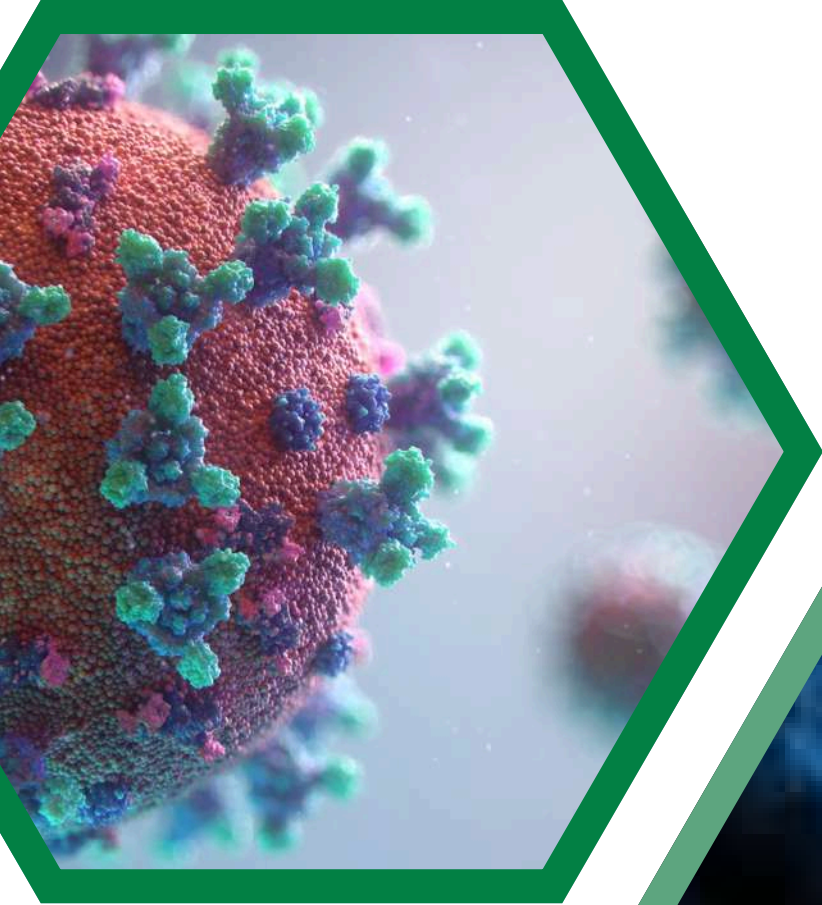




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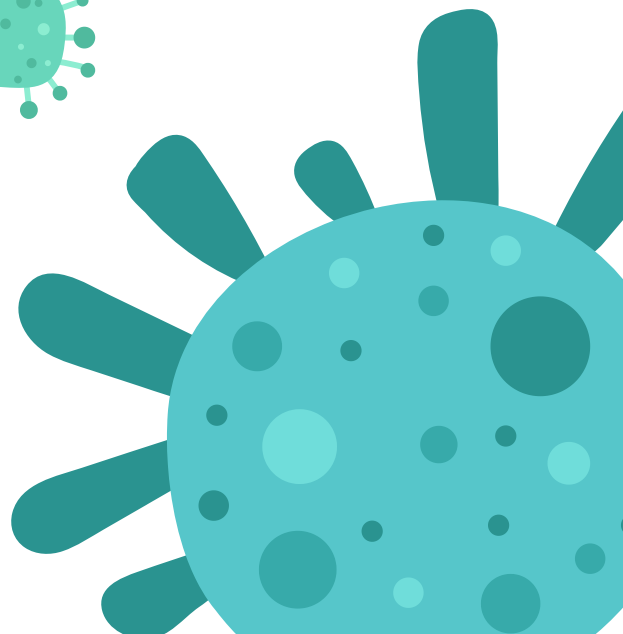


COVID-19

REPORT

AUDRY REFI MAULANA RIFQY

BIGGEST FENOMENAL FOR THE PAST 2 YEARS





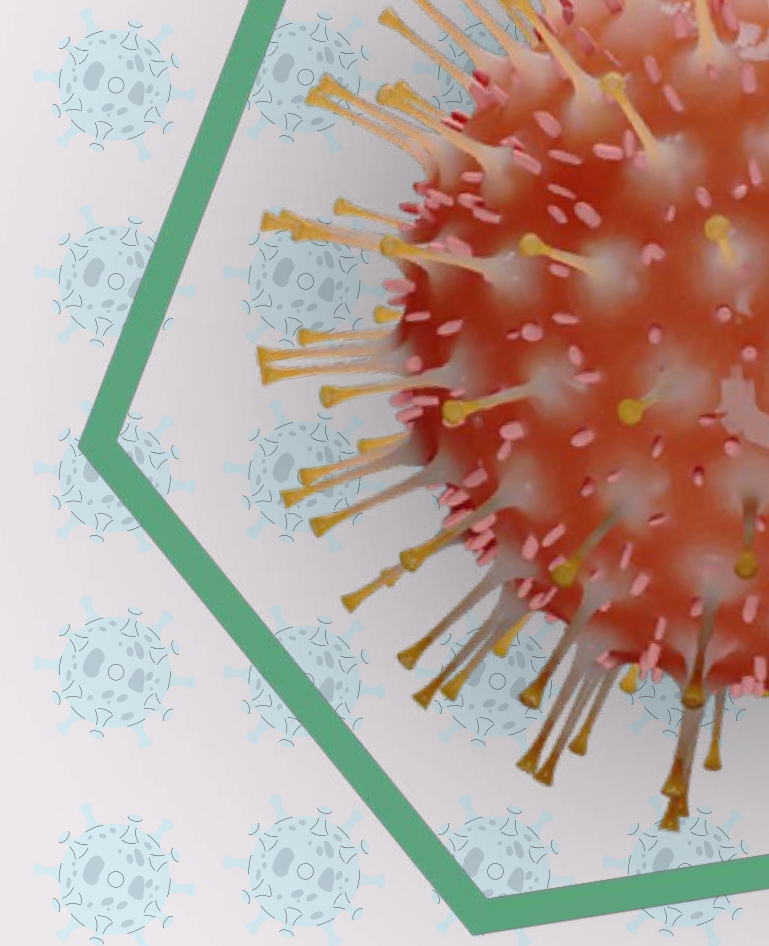
INTRODUCTION

1.1 Introduction of Covid 19

In the COVID-19 pandemic situation, data analysis becomes more important to understand trends and make decisions.

1.2 Problems Formulation

- Which Continents Are Affected the Most by Covid ?
- Which Countries Are Affected by the Most Covid ?
- Which Countries Have People Died Due to Covid ?
- Which Country Has the Most People's Deaths Due to Covid?
- Which Country Has the Most People Recovering From Covid?
- Which Country Has the Most People Recovering from Covid?





RESEARCH PURPOSES

2.1

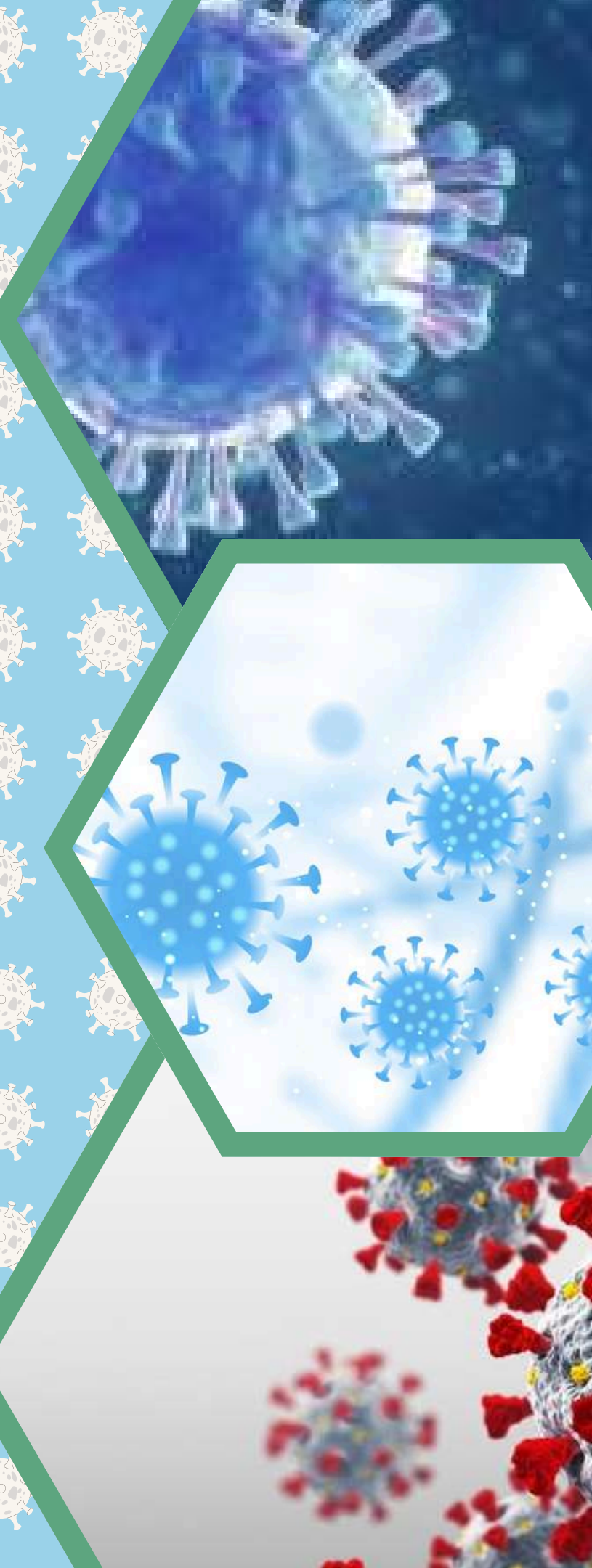
Understanding the Number of Countries With the Covid Virus Trapped Index

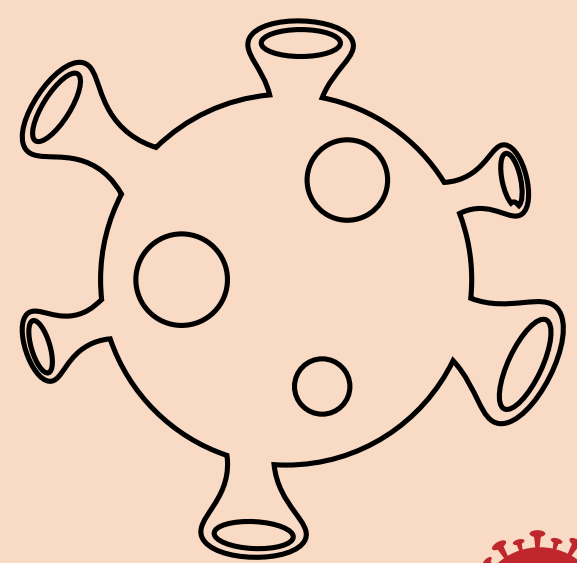
2.2

Understanding the Number of Countries From Each Continent That Have the Number of Deaths Due to Covid

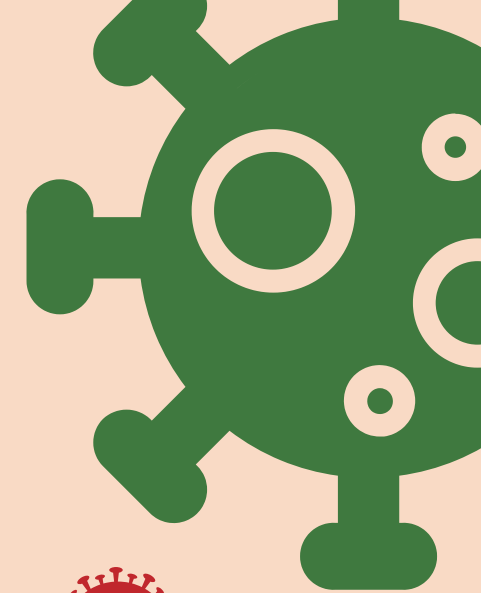
2.3

Understanding the Number of Countries From Each Continent That Have the Number of Deaths Due to Covid





RESEARCH PURPOSES

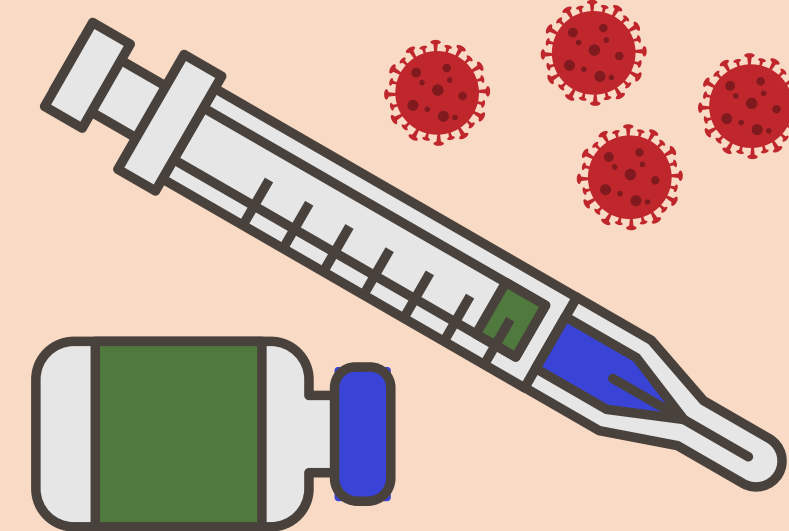
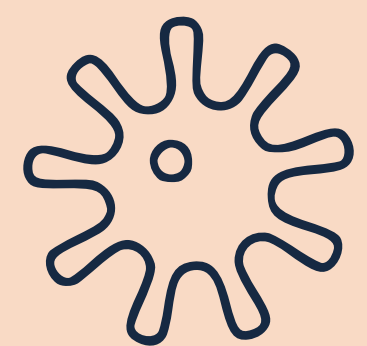
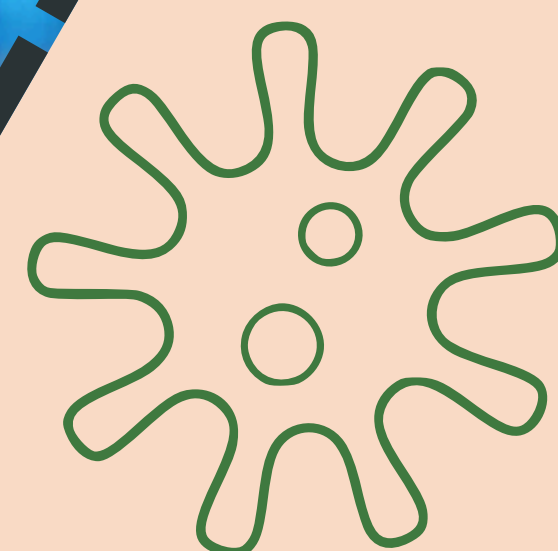


2.4 Understanding Which Countries Have The Highest Recent Cases Of Covid 19

2.5 Understanding Which Countries Have the Highest Recent Deaths From Covid 19

2.6 Understanding Which Countries Have the Highest Recent Recoveries From Covid 19

2.7 Understanding Which Countries Have the Highest Total of Recoveries From Covid 19

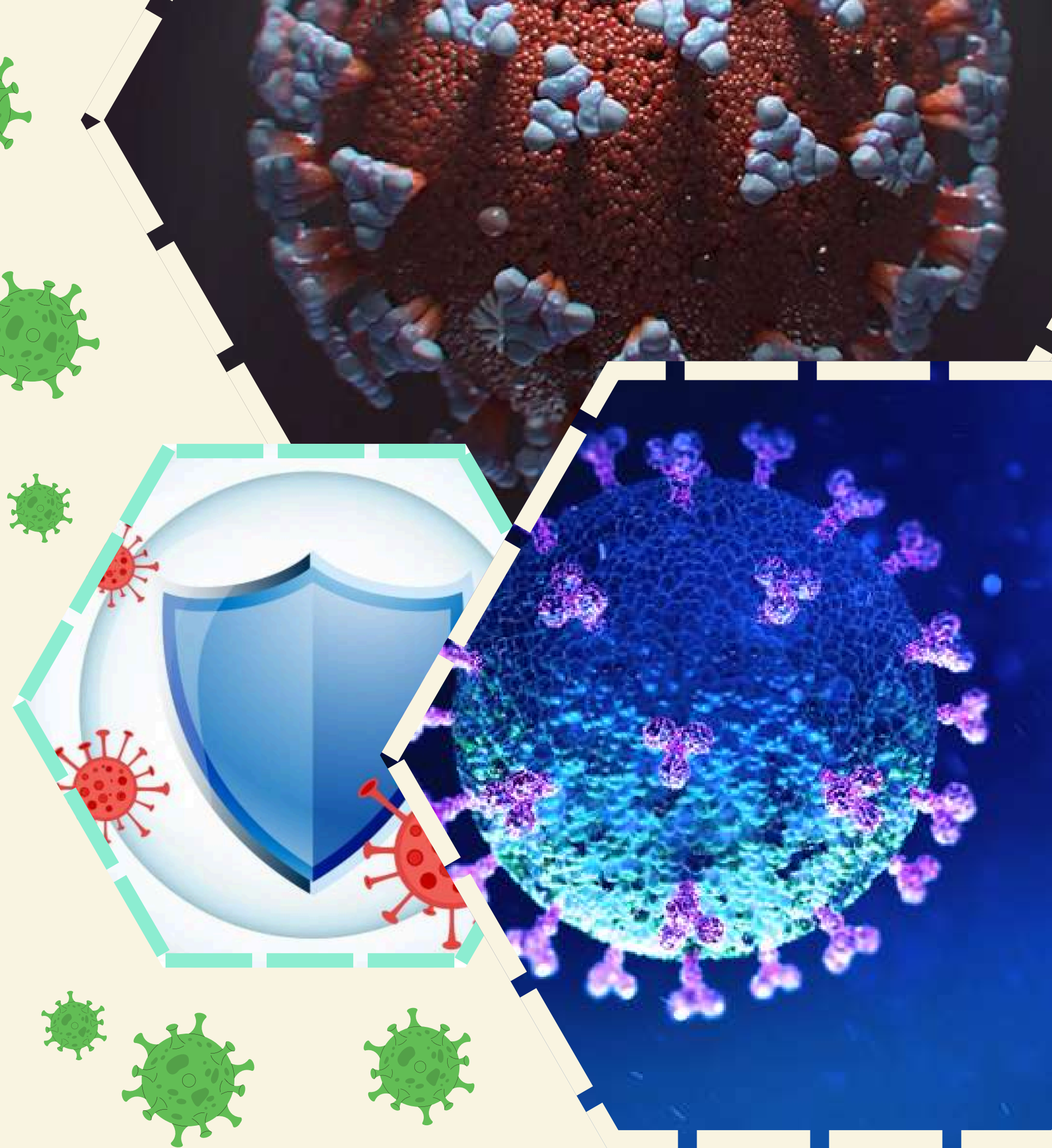




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RESEARCH PURPOSES

- 2.8** Understanding Which Countries Have the Highest Total of Deaths From Covid 19
- 2.9** Understanding the Country With 100 Cases of Recoveries From Covid
- 3.0** Understanding Countries With 100 Deaths From Covid



METODOLOGI PENELITIAN COVID

19

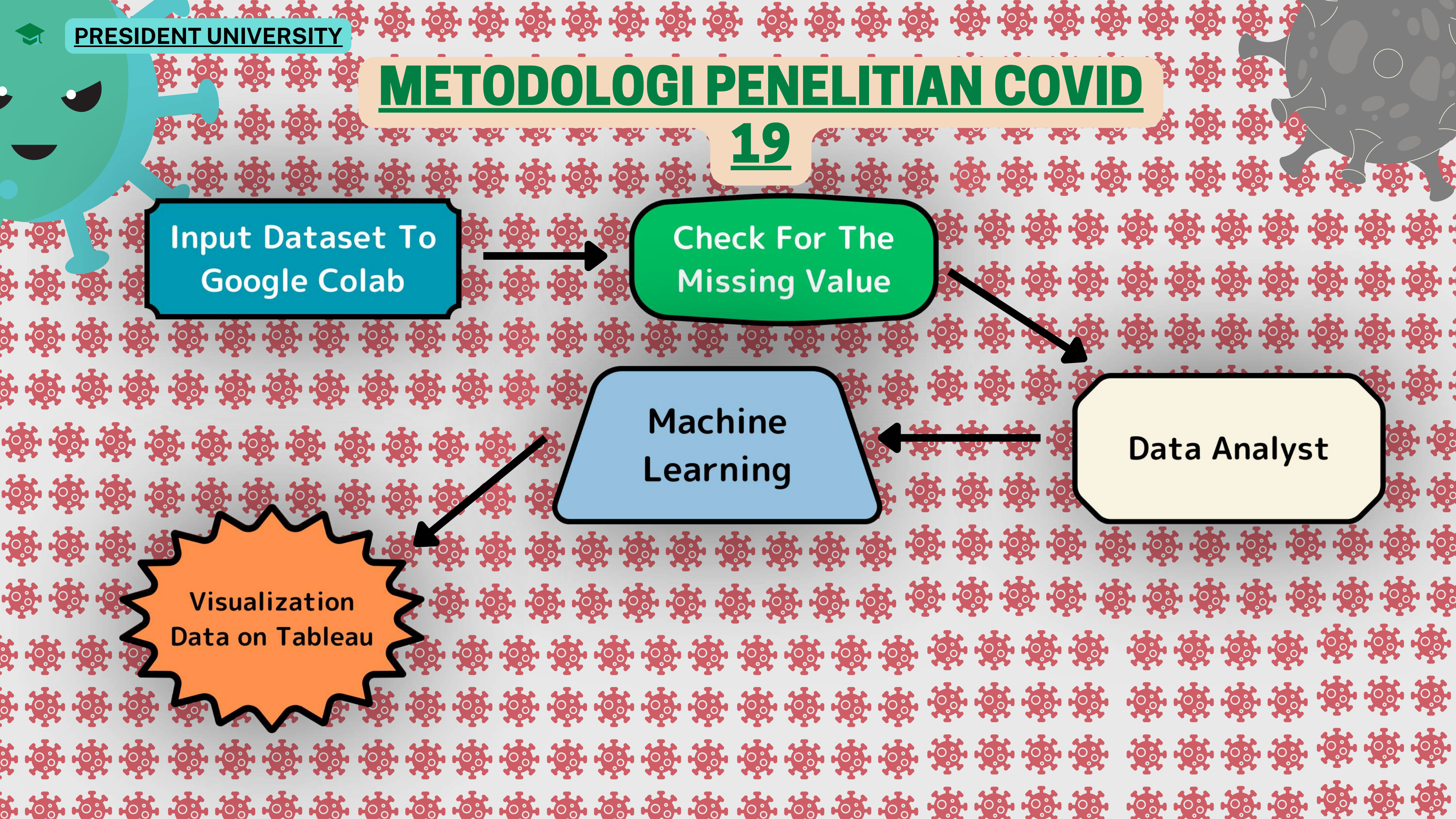
Input Dataset To
Google Colab

Check For The
Missing Value

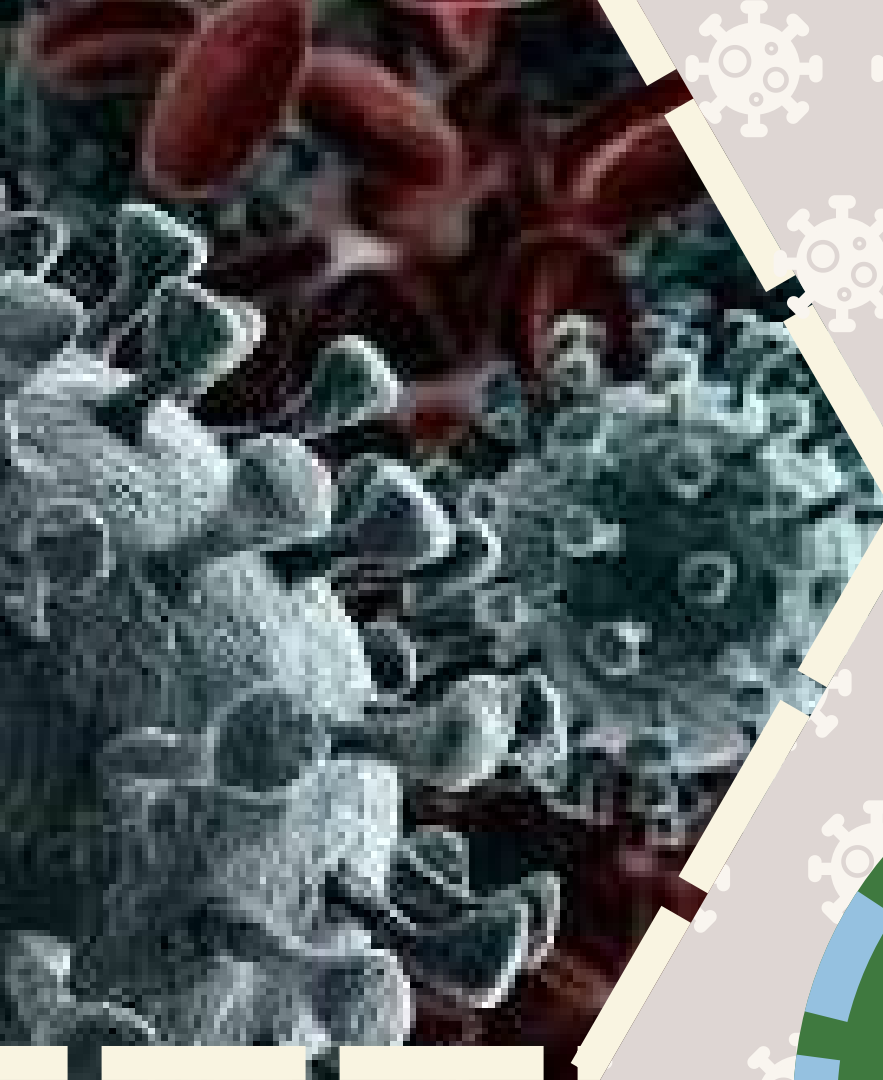
Machine
Learning

Data Analyst

Visualization
Data on Tableau



Data Preparation



IMPUTATION DATA

Imputation Data and Merging Data

Amount Of Data and Description Of Data

```
] copid.shape
```

```
(187, 15)
```

```
[ ] copid.describe()
```

	Confirmed	Deaths	Recovered	Active	New cases	New deaths	New recovered	Deaths / 100 Cases	Recovered / 100 Cases	Deaths / 100 Recovered	Confirmed last week	1 week change	1 week % increase
count	187.000000	187.000000	187.000000	187.000000	187.000000	187.000000	187.000000	187.000000	187.000000	187.00	187.000000	187.000000	187.000000
mean	43883.780749	1744.208556	29332.443850	13194.748663	411.42246	10.144385	329.288770	3.019519	64.820535	inf	40662.844920	2912.165775	13.606203
std	81764.215845	3977.697522	62404.530722	26278.092381	749.48395	23.897305	685.222177	3.454302	26.287694	NaN	76884.498993	5080.798764	24.509838
min	27.000000	0.000000	13.000000	2.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.00	24.000000	0.000000	-3.840000
25%	1114.000000	18.500000	626.500000	141.500000	4.000000	0.000000	0.000000	0.945000	48.770000	1.45	1051.500000	49.000000	2.775000
50%	5059.000000	108.000000	2815.000000	1600.000000	49.000000	1.000000	22.000000	2.150000	71.320000	3.62	5020.000000	432.000000	6.890000
75%	40460.500000	734.000000	22606.000000	9149.000000	419.500000	6.000000	221.000000	3.875000	86.885000	6.44	37080.500000	3172.000000	16.855000
max	293606.000000	15912.000000	241026.000000	98752.000000	2772.000000	96.000000	2613.000000	28.560000	100.000000	inf	276202.000000	18772.000000	226.320000

```
[ ] copid = pd.read_csv('/content/copid.csv')
```

```
[ ] copid.head()
```

	Country/Region	Confirmed	Deaths	Recovered	Active	New cases	New deaths	New recovered	Deaths / 100 Cases	Recovered / 100 Cases	Deaths / 100 Recovered	Confirmed last week	1 week change	1 week % increase	WHO Region
0	Afghanistan	36263	1269	25198	9796	106	10	18	3.50	69.49	5.04	35526	737	2.07	Eastern Mediterranean
1	Albania	4880	144	2745	1991	117	6	63	2.95	56.25	5.25	4171	709	17.00	Europe
2	Algeria	27973	1163	18837	7973	616	8	749	4.16	67.34	6.17	23691	4282	18.07	Africa
3	Andorra	907	52	803	52	10	0	0	5.73	88.53	6.48	884	23	2.60	Europe
4	Angola	950	41	242	667	18	1	0	4.32	25.47	16.94	749	201	26.84	Africa

DATA CLEANING

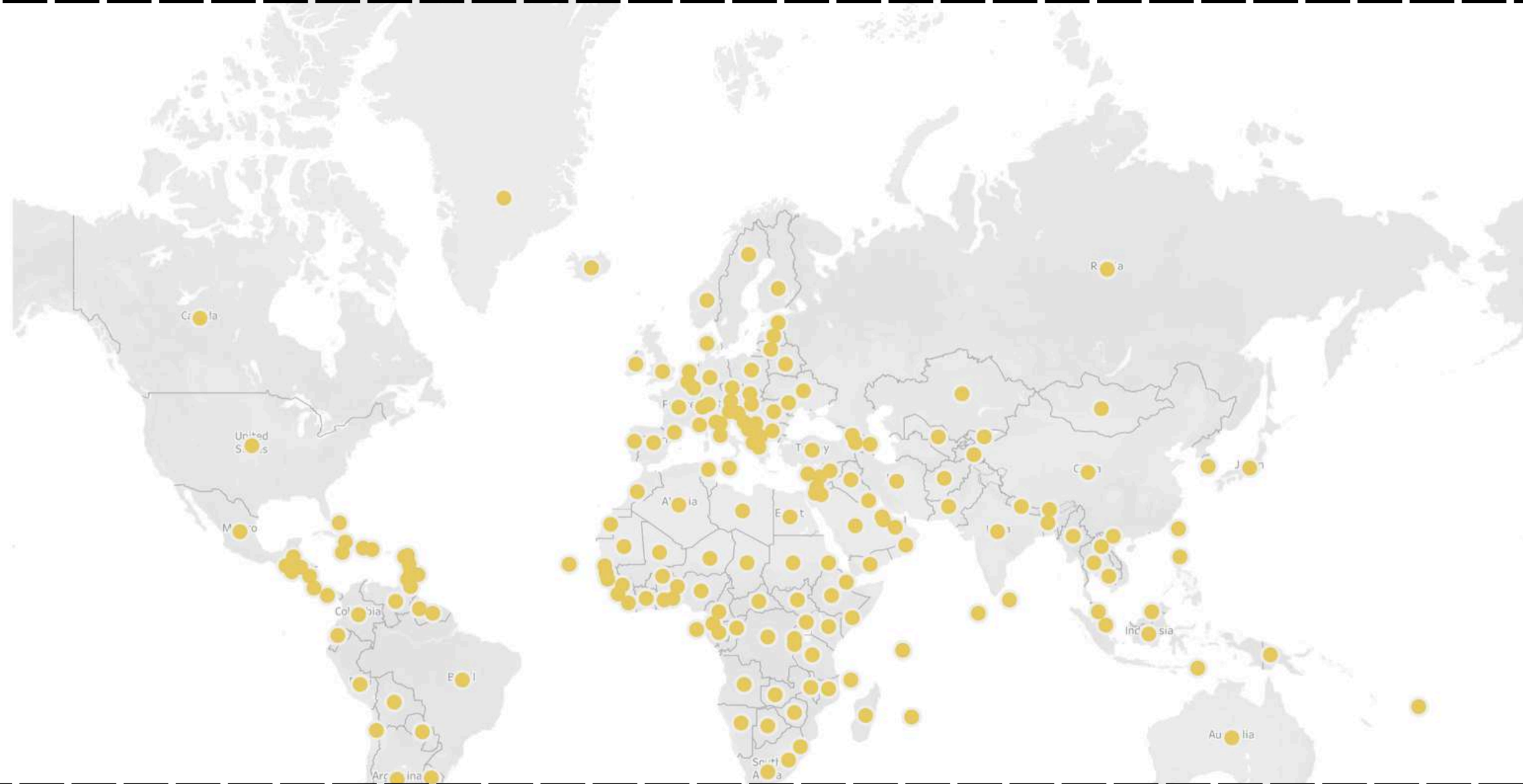
It looks like, there's no NaN, so we don't to clean that columns

Data columns (total 15 columns):

#	Column	Non-Null Count	Dtype
---	-----	-----	----
0	Country/Region	187 non-null	object
1	Confirmed	187 non-null	int64
2	Deaths	187 non-null	int64
3	Recovered	187 non-null	int64
4	Active	187 non-null	int64
5	New cases	187 non-null	int64
6	New deaths	187 non-null	int64
7	New recovered	187 non-null	int64
8	Deaths / 100 Cases	187 non-null	float64
9	Recovered / 100 Cases	187 non-null	float64
10	Deaths / 100 Recovered	187 non-null	float64
11	Confirmed last week	187 non-null	int64

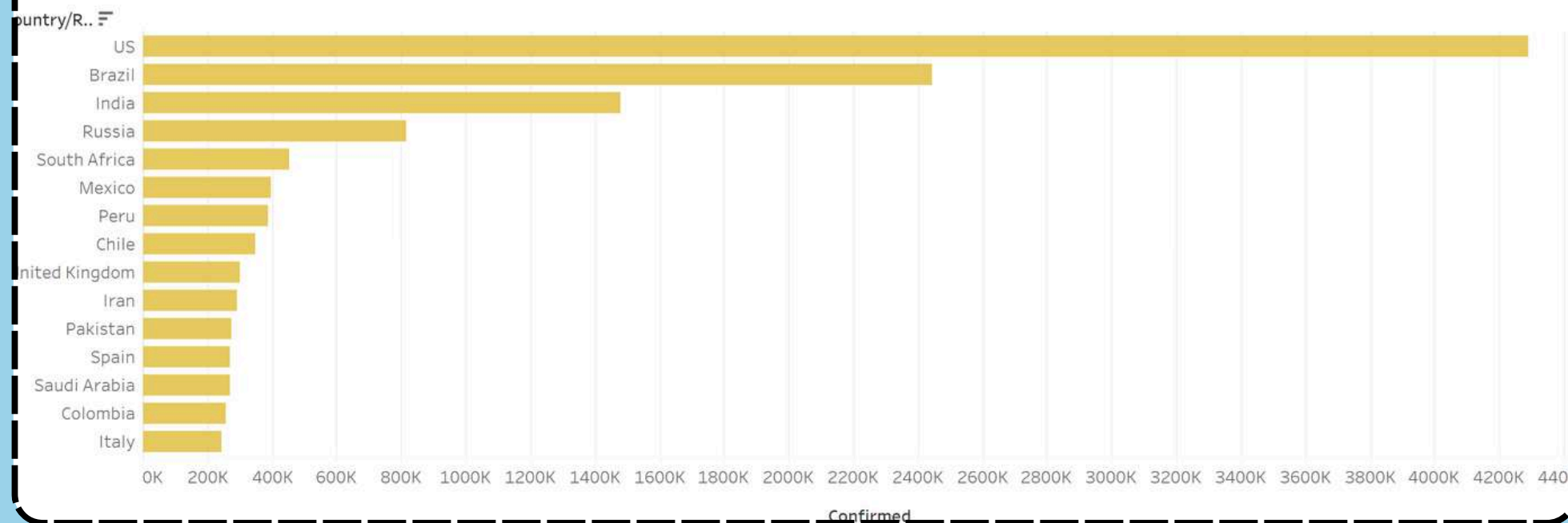
VISUALIZATION DATA

Visualization of Countries Affected by Covid Through Maps



Visualization of Countries Affected by Covid Through Maps

15 Countries That Have Confirmed Related to Covid 19

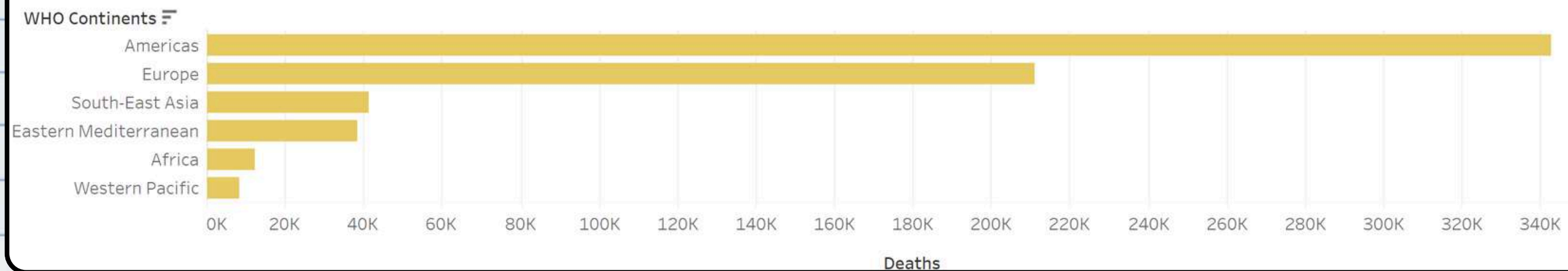


**Explanations:
Top 2 Countries
with the Highest
Number of
Confirmed
Cases:**

- **US: 4.3 million**
- **Brazil: 2.4 million**

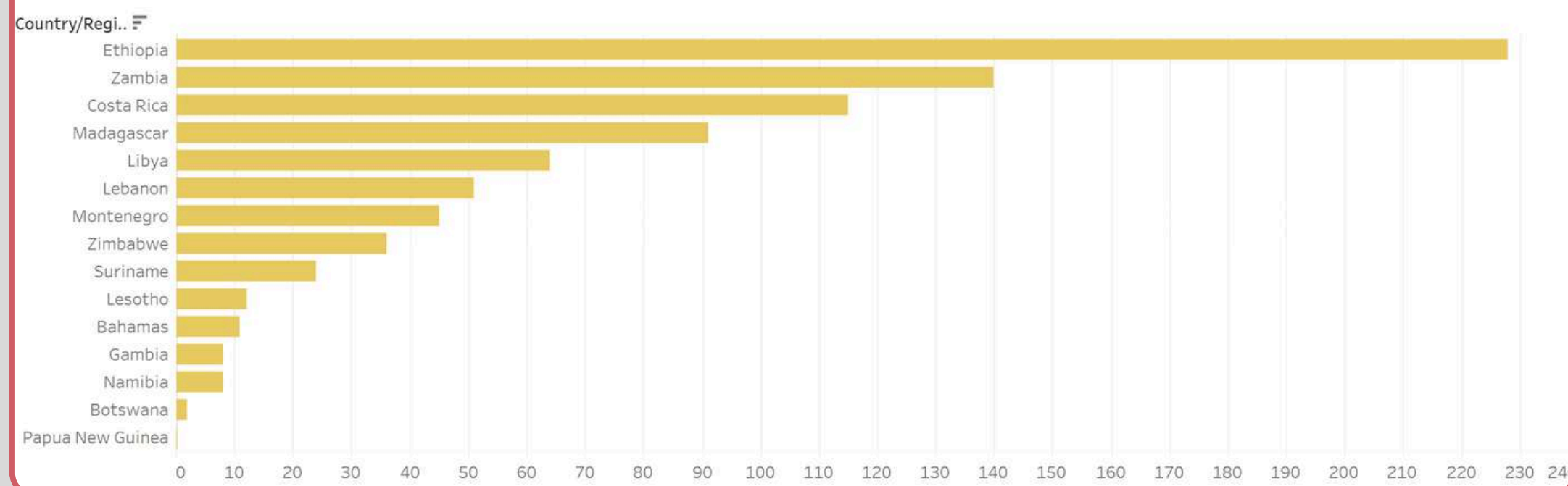
Continents with the highest number of deaths due to Covid

Continent with the highest number of deaths due to Covid

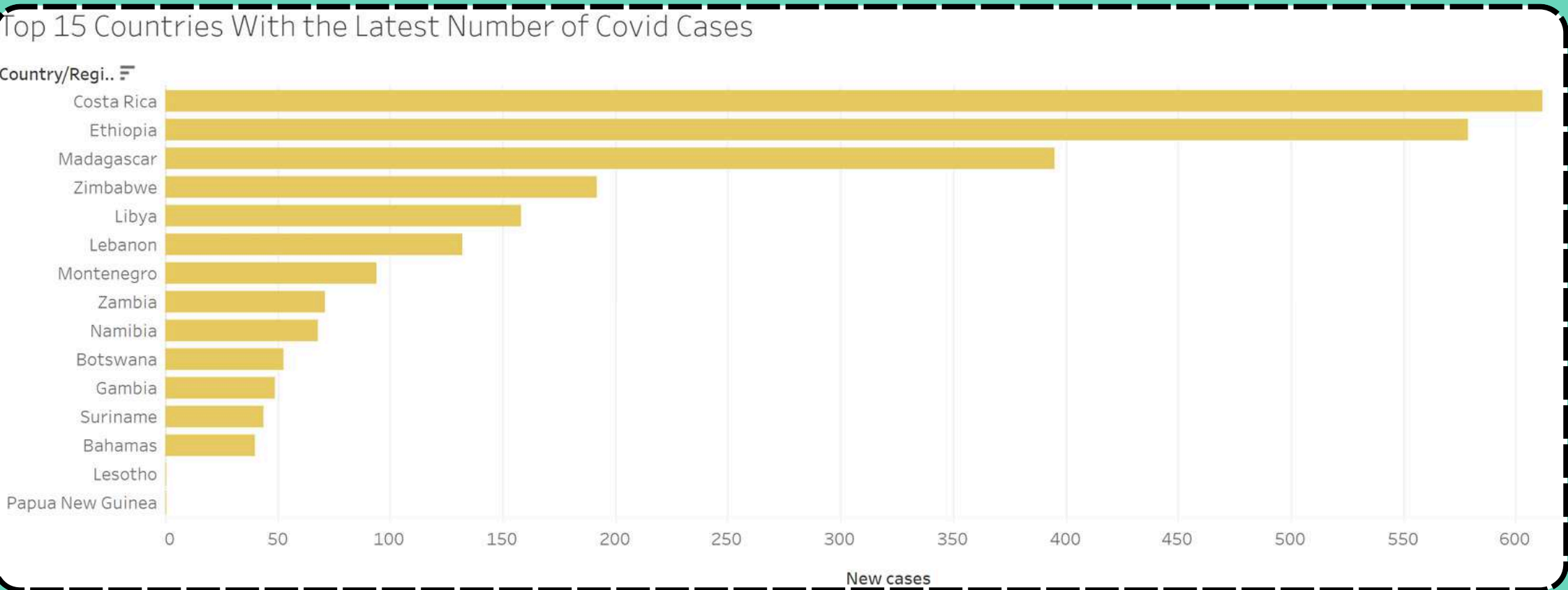


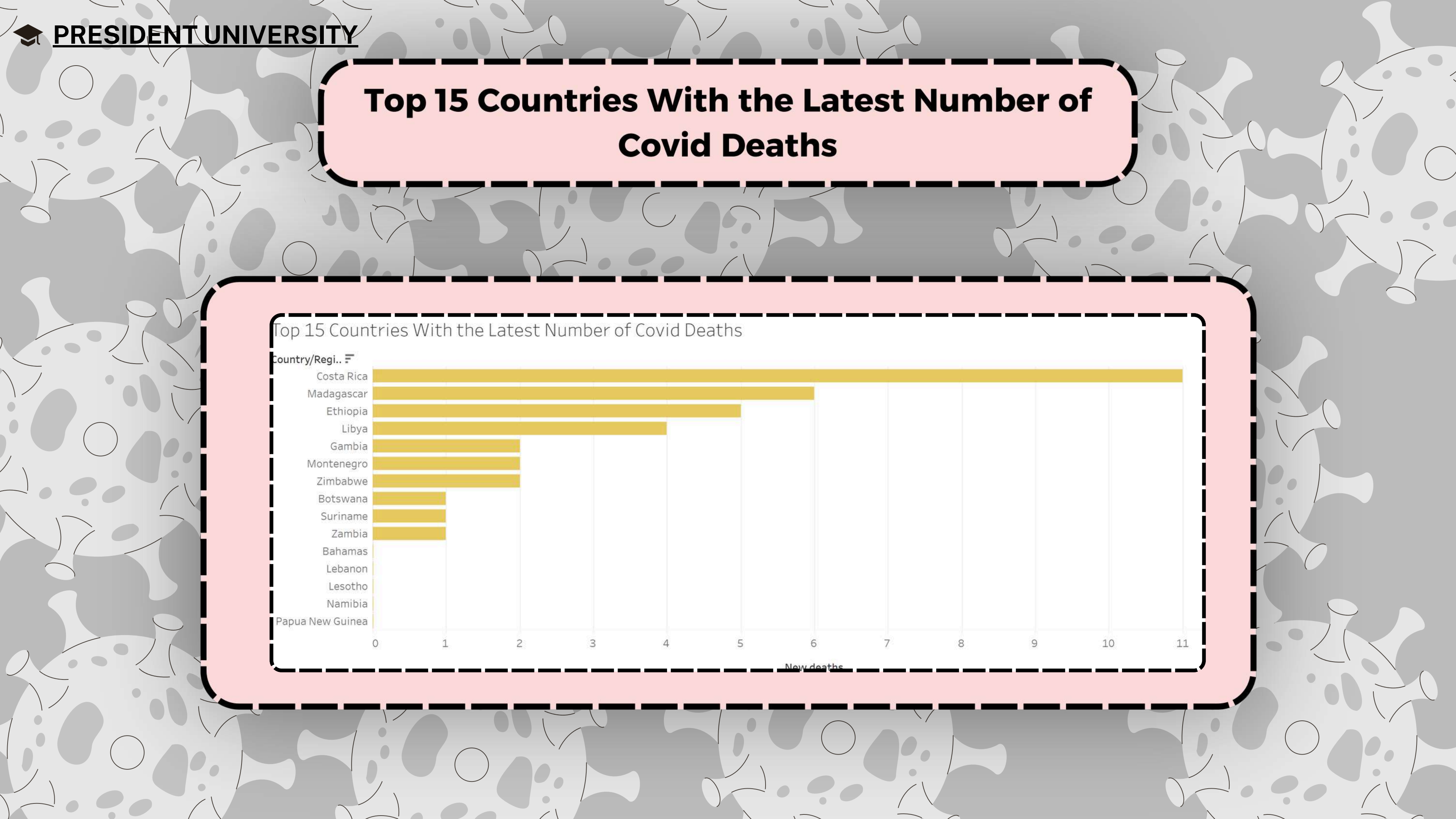
Top 15 Countries With the Most Deaths Due to Covid

Top 15 Countries With the Most Deaths Due to Covid

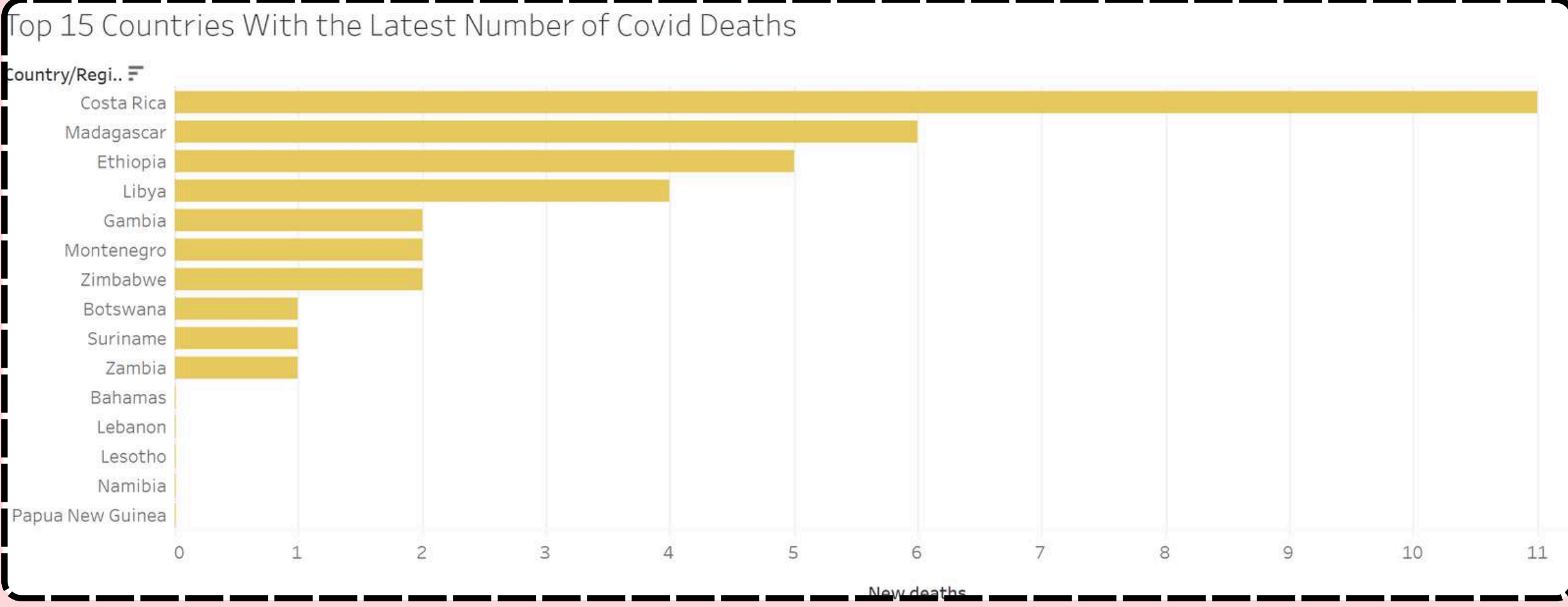


Top 15 Countries With the Latest Number of Covid Cases



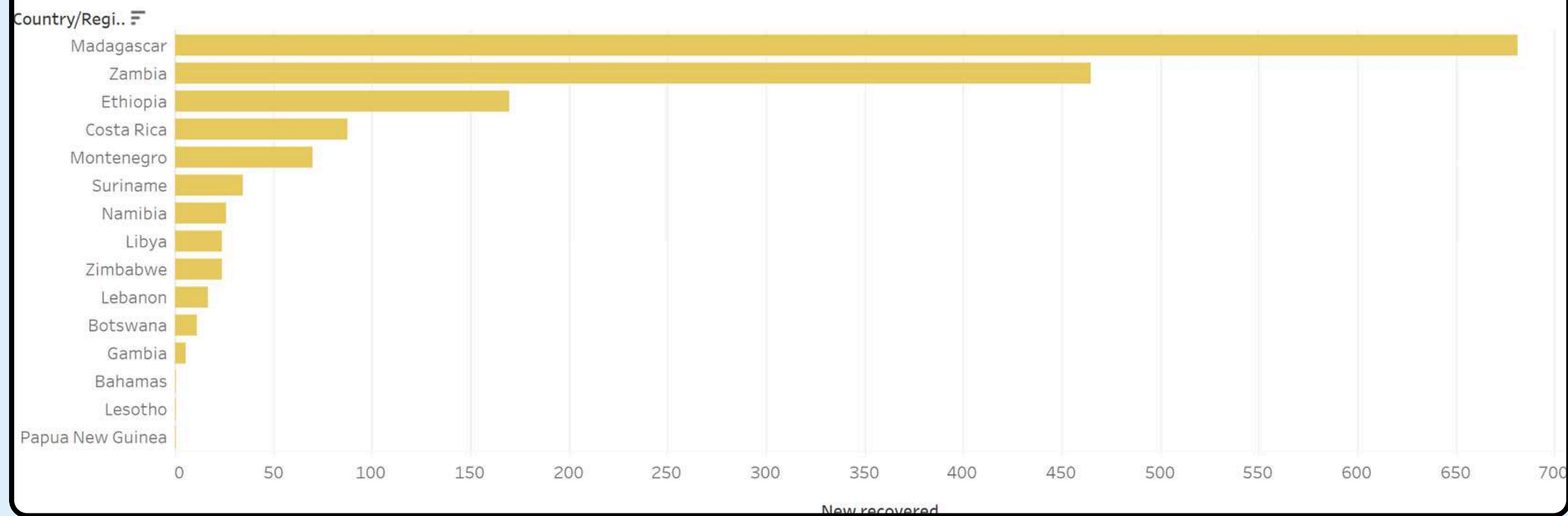


Top 15 Countries With the Latest Number of Covid Deaths



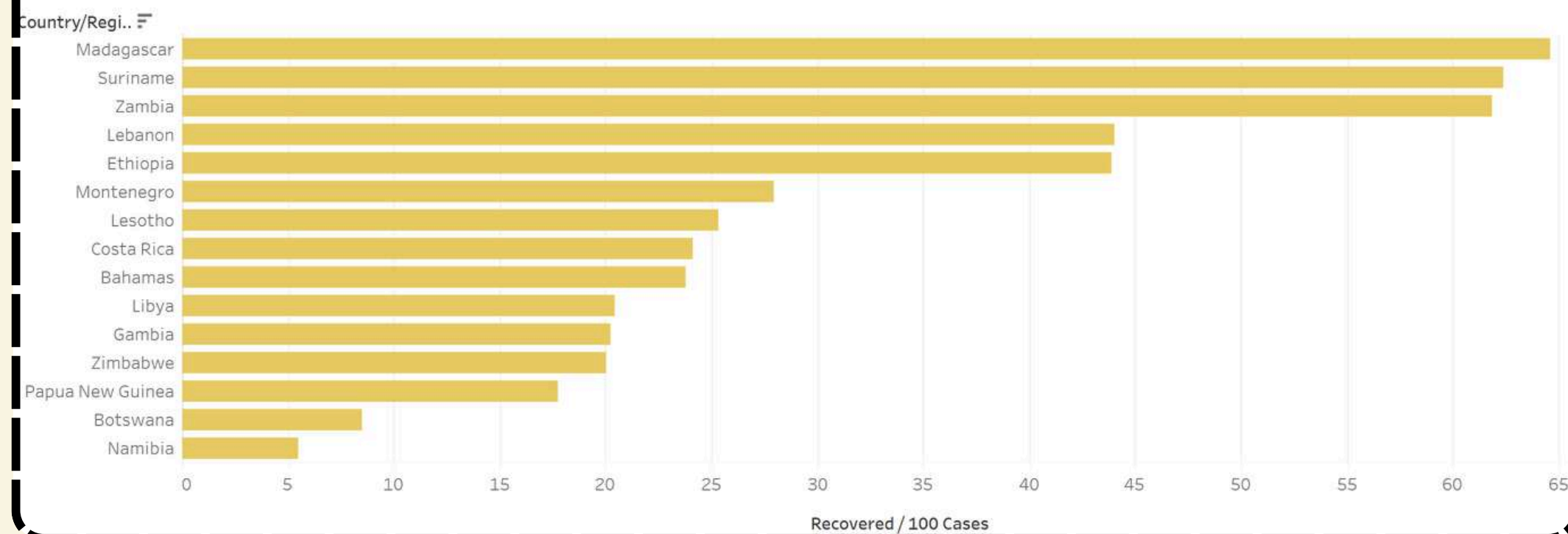
Top 15 Countries With the Latest Number of Covid Recoveries

Top 15 Countries With the Latest Number of Covid Recoveries



Top 15 Countries With 100 Covid Recoveries

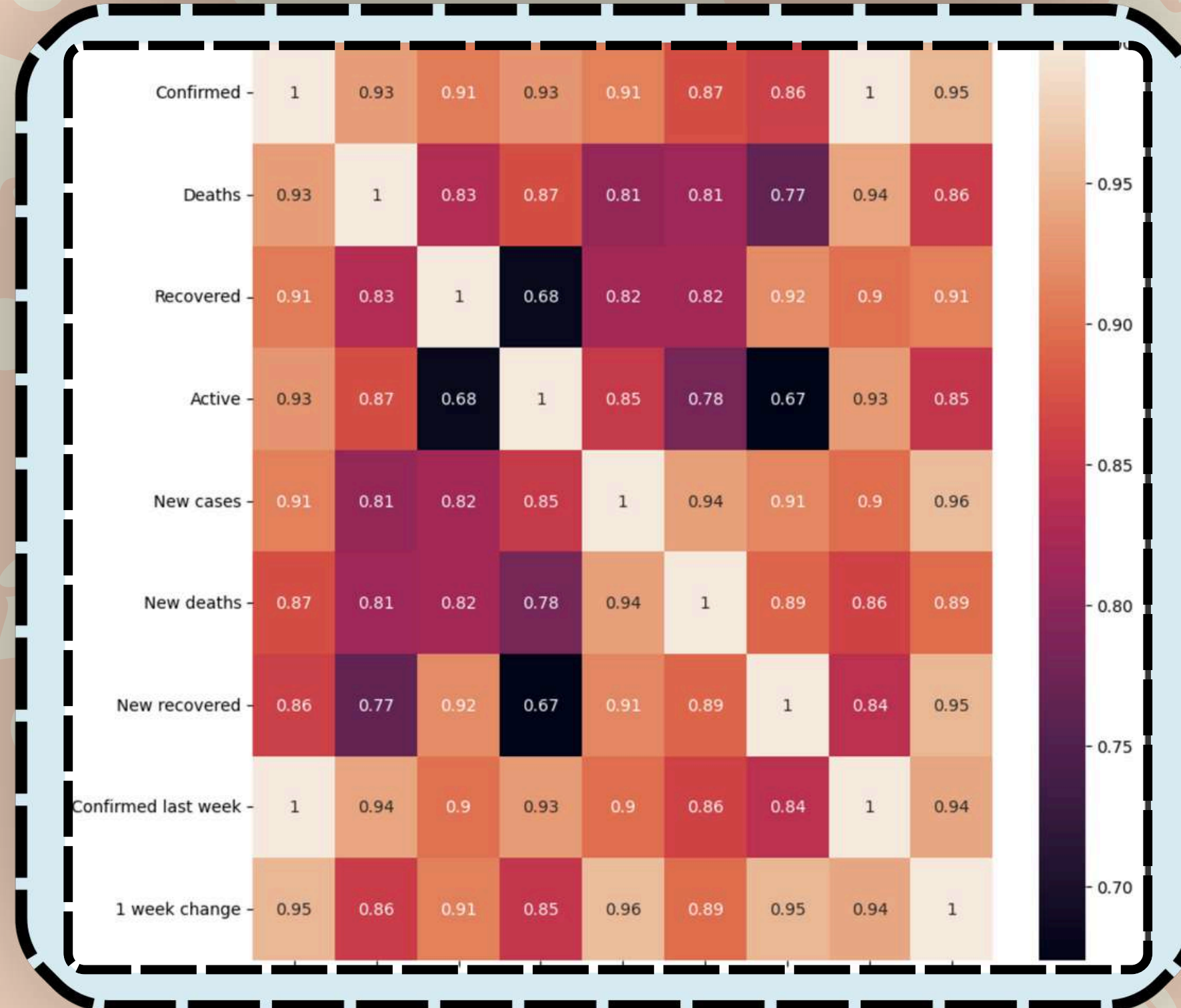
Top 15 Countries With 100 Covid Recoveries



Machine Learning

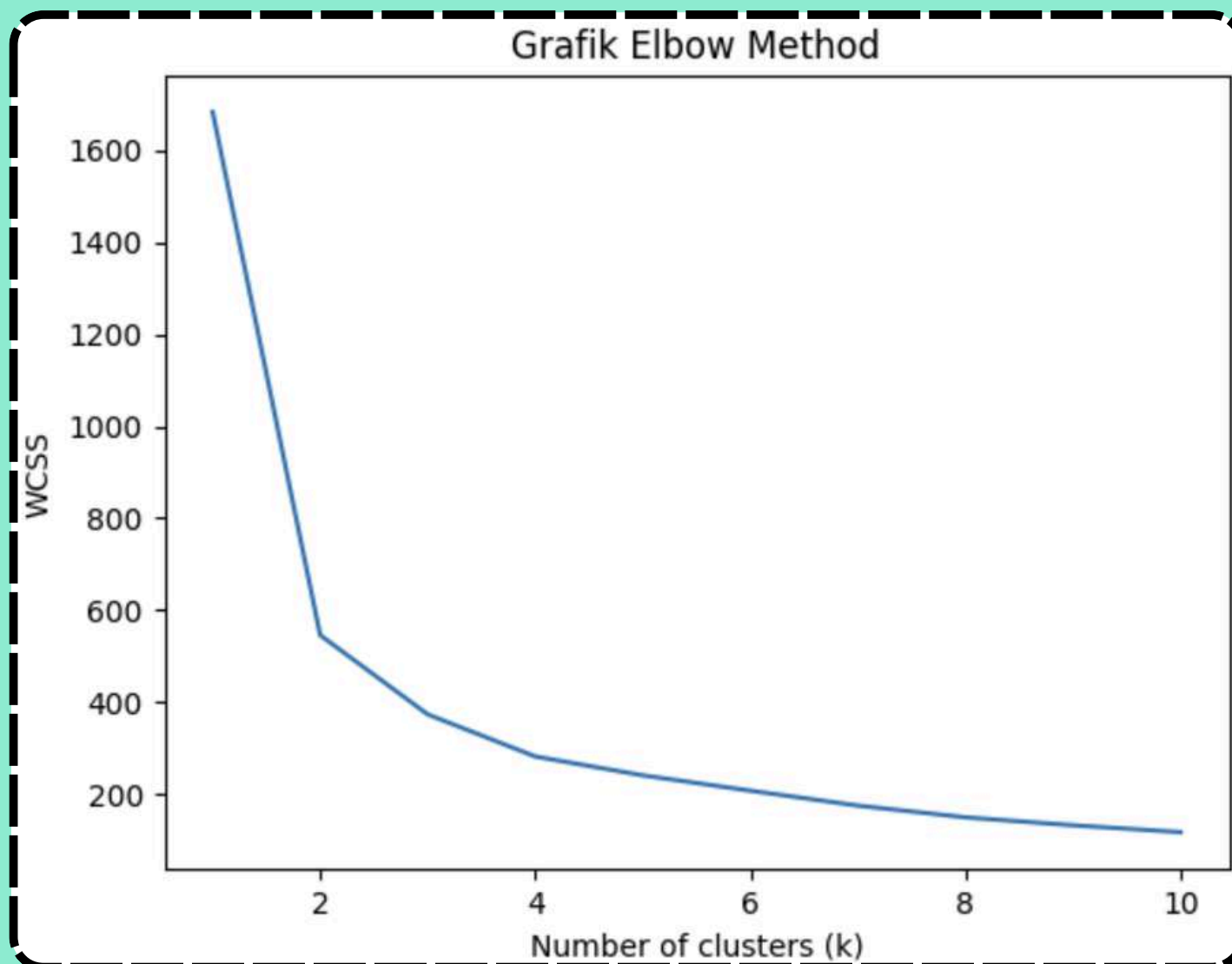


Correlation Matrix With Heatmap





CLUSTERING WITH K-MEANS





Insight and Recommendations

Insight:

- 'Confirmed' has a strong positive correlation with 'Deaths', 'Recovered', 'Active', 'New cases', and '1-week change' (above 0.9).
- 'New deaths' and 'New recovered' show a slightly lower, but still strong, positive correlation (around 0.87) with 'Confirmed'.

Recommendation:

- Focus on reducing transmission rates to control the spread of COVID-19, as it has a strong correlation with confirmed cases.
- Monitor 'New deaths' and 'New recovered' closely, as they show a slightly lower correlation with confirmed cases.

Insight of the K-Means Clustering Analysis:

- The new COVID-19 variant can cause significant changes in the distribution pattern, forming clearer and more separate groups of cases (clusters).
- Policy changes, such as lockdowns or vaccination campaigns, can drastically change the dynamics of the spread of the virus, forming new clusters.
- Seasonal factors can influence the increase and decrease of COVID-19 cases.

Recommendation of Machine Learning K-Means Clustering Analysis:

- Continuously monitor and adapt strategies to address emerging clusters and changing pandemic trends.
- Consider seasonal factors when developing policies to control the spread of COVID-19.



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**Thank you for the participation and
making time. See you again!**